

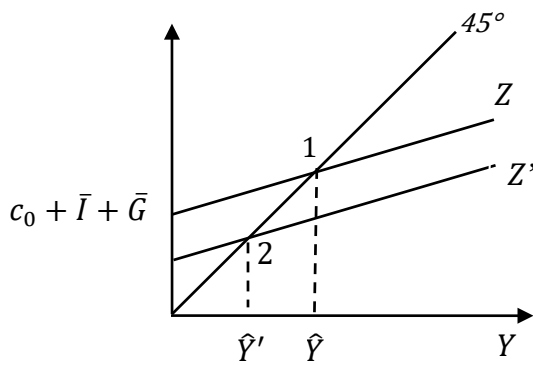
Question 1 (10 points)

The economy of a country in which only the goods market exists is described by the following system of equations:

$$\begin{aligned} C &= c_o \\ I &= \bar{I} + d_1 Y \\ G &= \bar{G} \\ T &= \bar{T} + tY \end{aligned}$$

where, as always, the positive constant c_o is autonomous consumption, t (a constant between zero and one) is the tax rate, $\bar{G}$ and $\bar{T}$ (both greater than zero) are the portions of government spending and net taxes that do not depend on income, and the parameter d_1 is positive, but smaller than one.

- a. Having determined graphically in the figure below the equilibrium level of production ($\hat{Y}$) for this economy, derive the analytical expressions of $\hat{Y}$, of autonomous spending A and of the multiplier implied by the model specified above.



In this economy, the demand for goods is

$$Z = c_o + \bar{I} + \bar{G} + d_1 Y,$$

where $c_o + \bar{I} + \bar{G}$ is autonomous spending, A . In the graph on the left, Z is therefore a straight line with slope d_1 , by assumption positive but smaller than one. Writing down the goods market equilibrium condition, and solving for Y , one gets the following expression for equilibrium income (the value of Y corresponding to point 1 in the figure):

$$\hat{Y} = \frac{c_o + \bar{I} + \bar{G}}{1 - d_1},$$

from which it follows that the multiplier is

$$\frac{1}{1 - d_1}.$$

- b. Suppose that the government cuts $\bar{T}$ at the same time that c_o falls by an identical amount, so that $\Delta \bar{T} = \Delta c_o < 0$. Show in the graph the effects of those changes and, using the results derived when answering the previous point of this question, derive the analytical expressions of the changes in equilibrium production, private saving, government saving and national saving caused by the decreases in $\bar{T}$ and in c_o . In your answer, make sure to discuss in each case why reaching a definite conclusion about the direction in which those variables will change is possible, or not possible.

When $\Delta \bar{T} = \Delta c_o < 0$, the aggregate demand curve in the figure shift downwards in a parallel fashion by the amount by which autonomous consumption falls, and the new equilibrium becomes point 2. From the expression for $\hat{Y}$ just derived, it follows that equilibrium income goes down by

$$\Delta \hat{Y} = \frac{1}{1 - d_1} \Delta c_o (< 0).$$

The change in private saving will be

$$\Delta \hat{S}_{priv} = \Delta[(\hat{Y} - \bar{T} - t\hat{Y}) - c_o] = (1 - t)\Delta \hat{Y} - \Delta \bar{T} - \Delta c_o = \frac{-1 - t + 2d_1}{1 - d_1} \Delta c_o,$$

a quantity whose sign depends on the relative size of the parameters t and d_1 , while government saving will fall by

$$\Delta \hat{S}_{gov} = \Delta \bar{T} + t\Delta \hat{Y} = \frac{1 - d_1 + t}{1 - d_1} \Delta c_o (< 0).$$

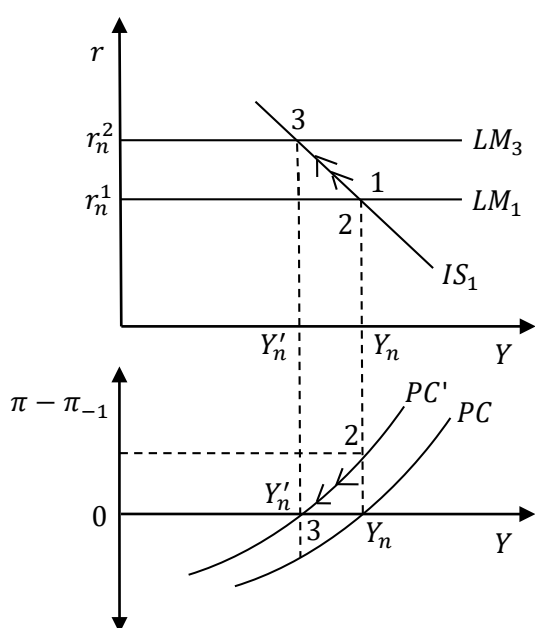
Finally, and although without more information it is not possible to determine the direction in which private saving will change, it is easy to conclude that national saving, and investment, will fall by

$$\Delta \hat{S}_{nat} = \Delta I = \frac{d_1}{1 - d_1} \Delta c_o.$$

Question 2 (10 points)

Consider country Macro, described by a standard IS-LM-PC model in which the central bank chooses the real interest rate and individuals form their expectations about the rate of inflation according to the rule $\pi^e = \pi_{-1}$.

- a. Suppose that, while it was in an initial medium run equilibrium that you will denote by '1' in the figure below, Macro is hit by an adverse supply shock consisting in a rise in the price of oil. Show in the graph the new short run equilibrium (to be denoted by '2') and the new medium run one ('3'). Finally, compare the composition of aggregate demand prevailing in the initial equilibrium '1' with that prevailing at '3'.



The rise in the price of oil will lead firms to increase the unit price that, for any given nominal wage, they charge for the goods they produce. In the WS-PS diagram the PS curve will shift downwards, the natural rate of unemployment will increase, and the associate natural level of output will therefore decrease. In the IS-LM-PC diagram, the PC curve shifts upwards. If the central bank does not change the policy rate, the new short-run equilibrium is point 2 in the two panels, where output is unchanged and inflation higher. The rise in inflation is due to the fact that, as discussed above firms now supply each unit of output at a higher price. Given last period's prices, a higher current price level implies more inflation. To prevent inflation from rising period after period, sooner or later the central bank will have to raise the policy rate. When this latter will be set to r_n^3 , the economy will once again be in a medium-run equilibrium position (point 3), with constant inflation and output at its (new) natural level. Compared to the initial equilibrium (1), in this new medium-run equilibrium (3) both consumption (due to the fall in Y) and investment (due to the fall in Y and the rise in r) are lower, while G is of course unchanged.

- b. To permanently return the economy to a medium run equilibrium with an income level Y_n equal to that prevailing before the adverse shock, economist Ann suggests to implement an expansionary monetary policy, while economist Margaret – to avoid the pick up in inflation that in her opinion would be the inevitable outcome of a looser monetary policy - suggests to resort to a fiscal expansion. Having shown in the graph the equilibria that would be reached following each of those suggestions, explain if, and why, you agree with Ann or with Margaret - or, in case you disagree with both, suggest a policy intervention that could permanently bring back the country to the natural level of output prevailing before the shock.

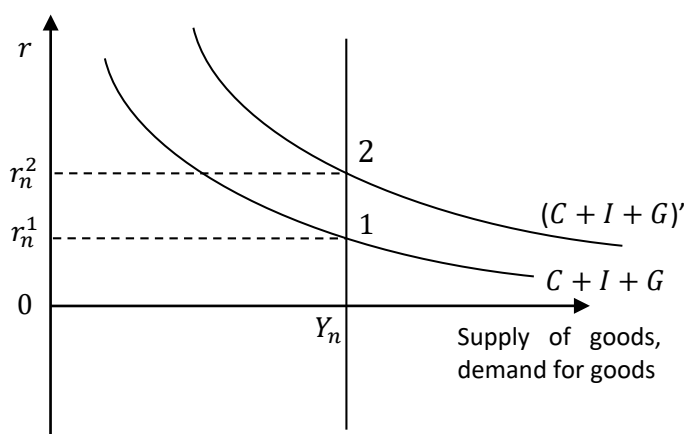
Since, due to the adverse shock, the initial level of production Y_n now exceeds the new natural level of output, monetary or fiscal policy aiming at keeping Y at Y_n , or in any case above Y'_n , would result in runaway inflation, and would therefore sooner or later force the central bank to allow the economy to return to the medium run equilibrium with $Y = Y'_n$. To permanently bring back income to Y_n , there is no alternative to the strategy consisting in making it once again the natural level of production, and this through the implementation of supply-side policies such as those leading to an increase in labor productivity, greater labor market flexibility, more competition in the goods markets, and the like.

Question 3 (10 points)

True or False?

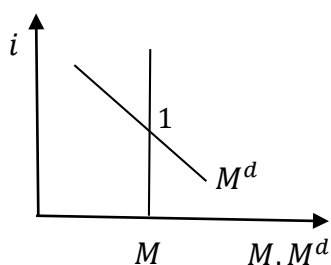
Explain whether the following statements are true or false. Motivate your answer in a brief but rigorous way, by making explicit reference to the relevant theory. Lack of proper explanations will result in zero points.

- a. Using the diagram “supply of goods-demand for goods” employed to analyse the determination of the real interest rate consistent with goods markets equilibrium in the medium run, r_n , explain if you agree, or do not agree, with the following statement: “In an economy in which consumption depends positively on disposable income, investment positively on Y and negatively on r , and in which government spending is exogenous, a change in consumers’ preferences leading to a rise in autonomous consumption is neutral in the medium run; in fact, it does not change the natural level of output, Y_n , as this latter depends on factors that have nothing to do with consumers’ preferences” [Hint: make sure to show in the graph below if and how the change under consideration affects the medium run equilibrium point at which the economy was located before the increase in autonomous consumption].



False. If c_0 rises, aggregate demand will be higher for any level of the real interest rate, and the demand curve in the figure will therefore shift to the right. The new medium run equilibrium becomes point 2. In the move from the initial equilibrium 1 to the new equilibrium 2, r_n rises by the amount needed to lower investment exactly by the amount by which autonomous consumption has gone up. It follows that, even though it does not lead to a change in the natural level of output, which depends on supply-side factors only, or in the level of the aggregate demand for goods, the change in c_0 is not neutral in the medium run.

- b. In country Gamma, the central bank chooses, and wants to keep constant at the chosen level, the interest rate, while in country Delta the central bank chooses the money supply (and, once again, keeps it constant at the level it has chosen). The figure below depicts the money market equilibrium in the two countries. As you can see, in this initial equilibrium position money supply is the same in both, and the same is true about the money demand curve and the equilibrium interest rate. Explain if you agree, or do not agree, with the following statement: “If, in both countries, banks decide to decrease by the same amount the fraction of deposits held as reserves, to keep the interest rate at the chosen level the central bank of Gamma will have to increase the money supply, while in Delta – whose central bank wants to prevent the money supply from changing – the interest rate will fall”. [Hint: in the figure, denote by 2_γ and 2_δ the new equilibria, if any, that will be reached by Gamma and Delta, respectively].



False. The equilibrium position will remain at point 1 in both countries. In fact, the fall in the reserve ratio θ leads to an increase in money supply (banks will lend out a larger fraction of any given amount of deposits received by their customers, thus causing a further increase in deposits, bank loans, deposits ... and therefore M).

- In Gamma, the central bank must bring back M to its initial level (e.g., by selling bonds in the open market), and this to prevent the interest rate from falling below the chosen level;
- in Delta, the central bank must do the same, in this case because it wants keep M constant at the chosen level.

While shocks to money demand affect the money market equilibrium in a way that depends on whether the central bank is choosing, and keeping constant, the money supply rather than the interest rate, the effects of shocks to the money supply do not depend on the rule of conduct followed by the central bank.